

UNIVERSITY OF GHANA

DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE

ASTAA 607_COMPUTATIONAL STATISTICS 1

ID22424240

1a)

```
### Q1.a). Describing the crabs dataset  
?crabs
```

Output:

a. Describe the dataset using the one in package MASS.

```
> library(MASS)
```

```
> # Question 1
```

```
> # a. Describe the dataset using the one in package MASS.
```

```
> library(MASS)
```

Description

The crabs data frame has 200 rows and 8 columns, describing 5 morphological measurements on 50 crabs each of two colour forms and both sexes, of the species *Leptograpsus variegatus* collected at Fremantle, W. Australia.

Usage

crabs

Format

This data frame contains the following columns:

sp

species - "B" or "O" for blue or orange.

sex

as it says.

index

index 1:50 within each of the four groups.

FL

frontal lobe size (mm).

RW

rear width (mm).

CL

carapace length (mm).

CW

carapace width (mm).

BD

body depth (mm).

Source

1b)

```
### ### Q1. b. Load the dataset and display its structure  
data("b")
```

Output:

Q1. b)

str(crabs) # For checking the structure of the dataset2 ### Q1. b) Displaying the stricture of the dataset# b. Load the dataset and display its structure

```
> data("crabs")
```

```
> # b. Load the dataset and display its structure
```

```
> data("crabs")
```

```
> data(crabs) ## To load the dataset, optional for crabs since crabs is already available in our R environment.
```

```
> ### Q1. b) Displaying the structure of the dataset
```

1c)

```
### Q1 c). # c. Produce the frequency table for species and sex  
table(crabs$sp, crabs$sex)
```

```
# c. Produce the frequency table for species and sex
```

```
> table(crabs$sp, crabs$sex)
```

```
F M
```

```
B 50 50
```

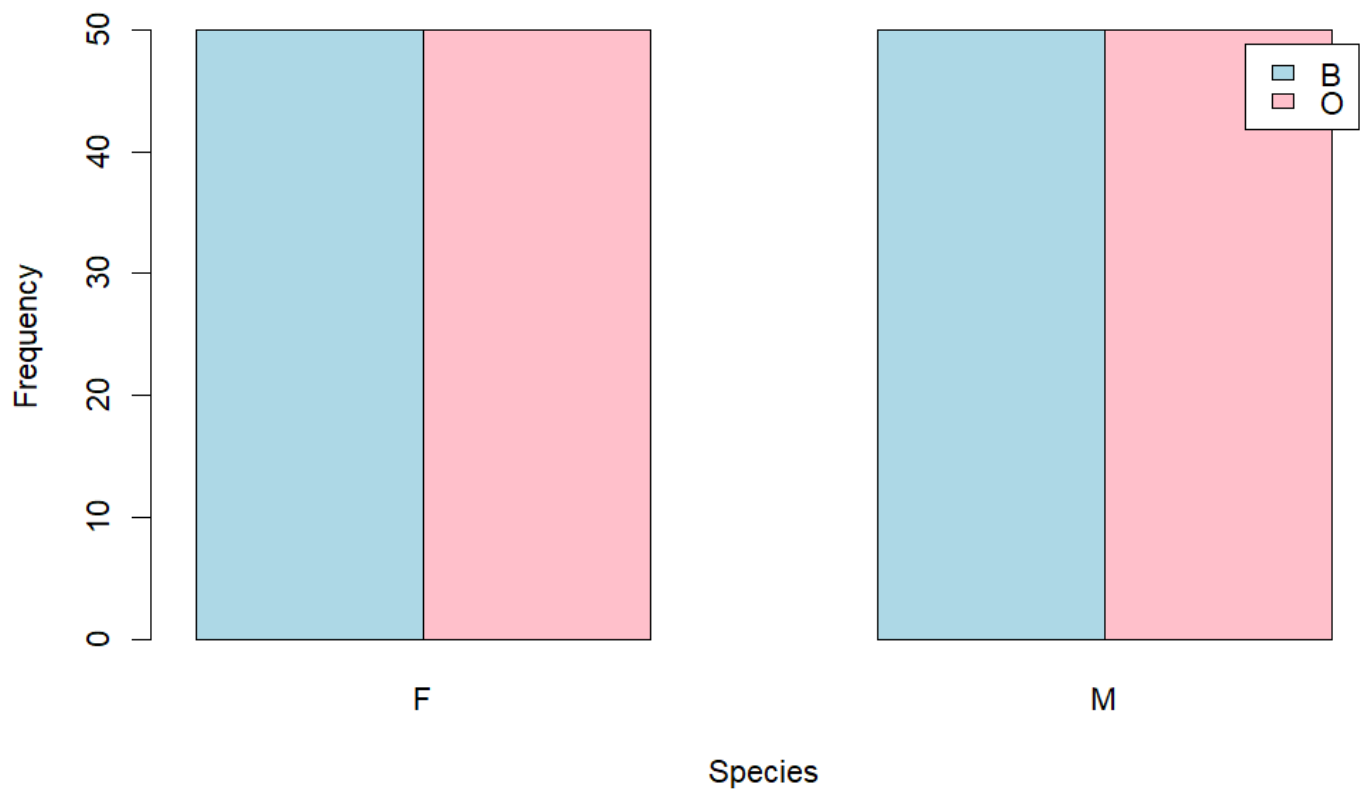
```
O 50 50
```

```
### Q1
```

```
    d. Construct an appropriate bar chart or mosaic plot  
    summarising the group composition.
```

```
barplot(table(crabs$crabs$sex),  
        beside = TRUE,  
        col = c("lightblue", "pink"),  
        legend = TRUE,
```

Species and Sex Distribution



"set"

```
Comment on whether the dataset is balanced across the four groups
```

```
# Yes – the dataset is perfectly balanced. 50 observations in each of  
the four groups
```

```
# Equal numbers across both species
```

```
# e. Comment on whether the dataset is balanced across the four groups
```

```
# Yes – the dataset is perfectly balanced. 50 observations in each of  
the four groups
```

1e# Question 2 - Distribution of Body Depth

a. Construct side-by-side boxplots of body depth (BD) for the four groups defined by species and sex

```
group <- interaction(crabs$sp, crabs$sex)
```

Output: # Question 2 - Distribution of Body Depth

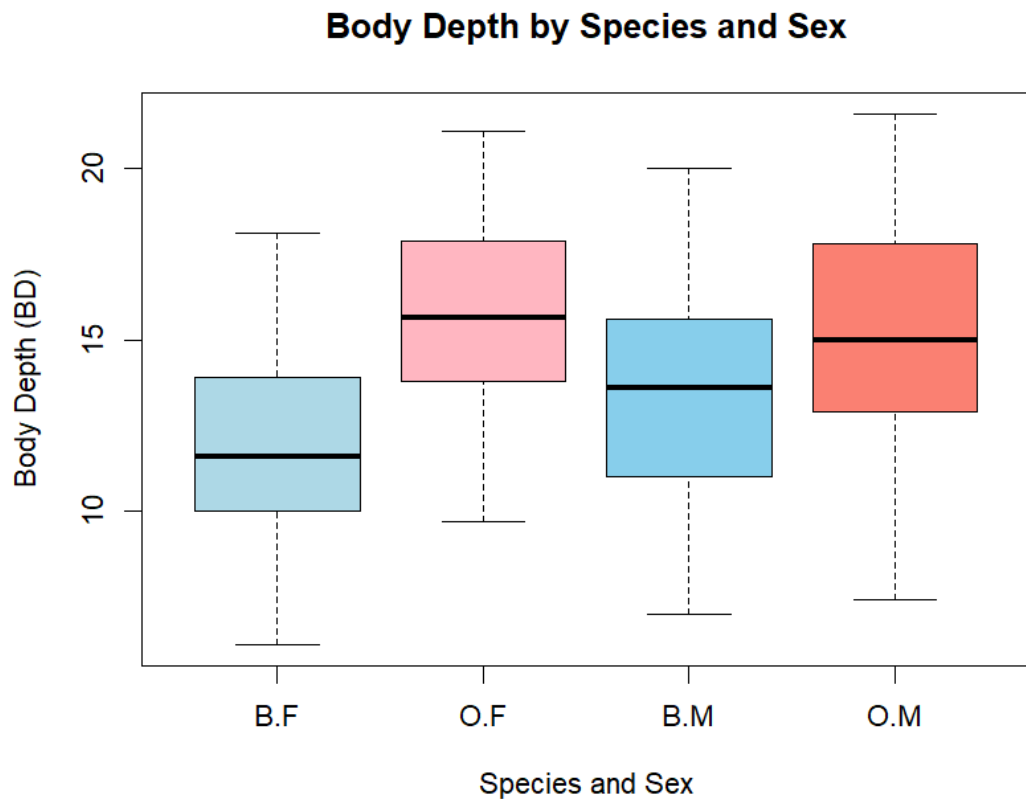
> # a. Construct side-by-side boxplots of body depth (BD) for the four groups defined by species and sex

```
> group <- interaction(crabs$sp, crabs$sex)
```

2b)

```
### Q1. # b. Identify any outliers and compare the central  
tendency and variability between groups  
by(crabs$BD, group, boxplot.stats) # there are no outliers
```

Output:



1)

```
### Q3 Question 3 - Carapace Length Distribution by Species
# a. Produce histograms of carapace length (CL) for each species
# Set plotting area to 1 row, 2 columns
par(mfrow = c(1, 2))
```

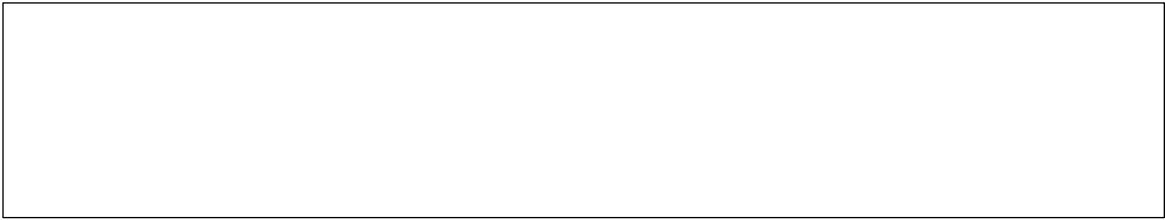
Output

1b) Question 3 - Carapace Length Distribution by Species

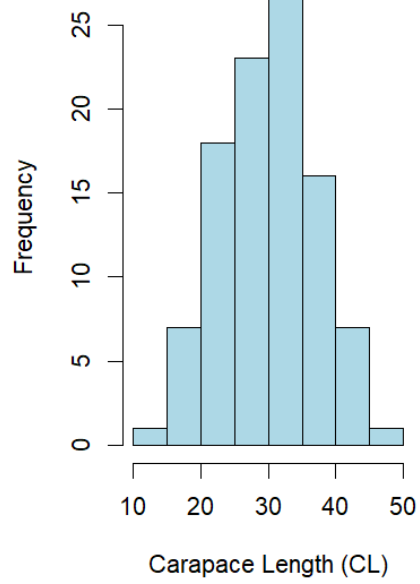
```
> # a. Produce histograms of carapace length (CL) for each species
```

```
> # Set plotting area to 1 row, 2 columns
```

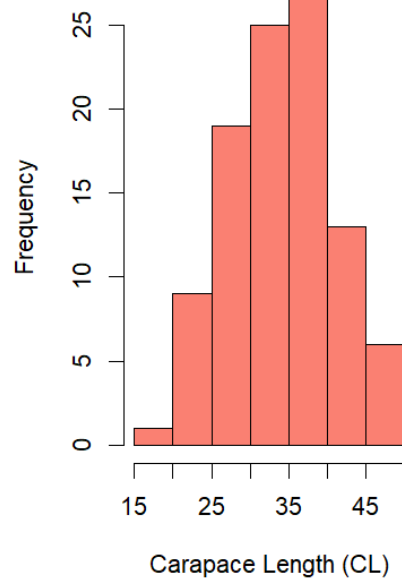
```
> par(mfrow = c(1, 2))
```



Carapace Length - Blue



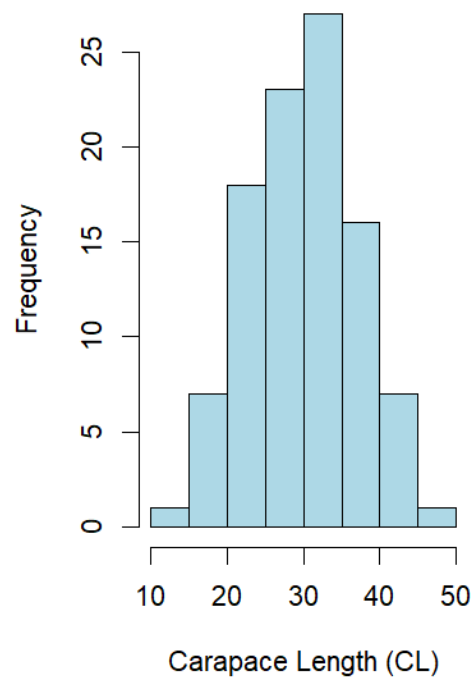
Carapace Length - Orange



Outp

ut:

Carapace Length - Blue



3b)

```
###  
# Orange species  
hist(crabs$CL[crabs$sp == "O"],  
      col = "salmon",  
      main = "Carapace Length - Orange",  
      xlab = "Carapace Length (CL)",  
      breaks = 10)
```

Output

3b)

Output:

1b)

```
b. Describe and compute the shapes of the distributions,  
commenting on skewness and modality
```

```
# - Shape and Modality
```

Output # Both species show approximately unimodal distributions.
The histograms are fairly smooth without multiple distinct peaks.

3b) # This suggests one main population per species

```
Question 4. - Comparison of Mean Measurements
```

Output # a. Compute the mean of each morphological measurements
(FL, RW, CL, CW, BD) by species and sex

```
means <- aggregate(cbind(FL, RW, CL, CW, BD) ~ sp + sex,  
                    data = crabs,
```

)

```
# Question 4. - Comparison of Mean Measurements
```

```
> # a. Compute the mean of each morphological measurements  
(FL, RW, CL, CW, BD) by species and sex
```

Output > means <- aggregate(cbind(FL, RW, CL, CW, BD) ~ sp + sex,
+ data = crabs,
+ FUN = mean)

4b)

```
# b. Construct a grouped bar chart to display these means  
# converting to matrix form (remove grouping columns)  
mean_matrix <- as.matrix(means[, 3:7])
```

Output: b. Construct a grouped bar chart to display these means

FUN = mean)

> # b.



Construct a grouped bar chart to display these means

> # converting to matrix form (remove grouping columns)

> mean_matrix <- as.matrix(means[, 3:7])

> # converting to matrix form (remove grouping columns)

> mean_matrix <- as.matrix(means[, 3:7])

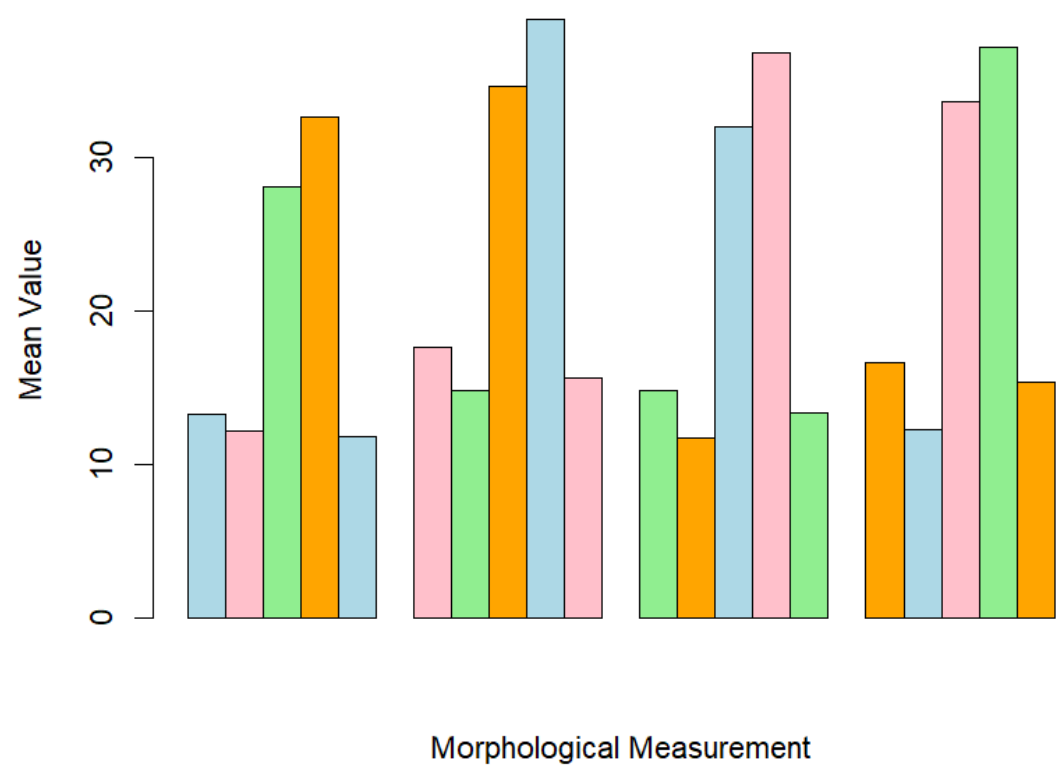
4b)

Output

```
# Grouped bar chart
barplot(t(mean_matrix),
        beside = TRUE,
        col = c("lightblue", "pink", "lightgreen", "orange"),
        legend = rownames(mean_matrix),
        xlab = "Morphological Measurement",
        ylab = "Mean Value",
        main = "Mean Morphological Measurements by Species
and Sex")
```



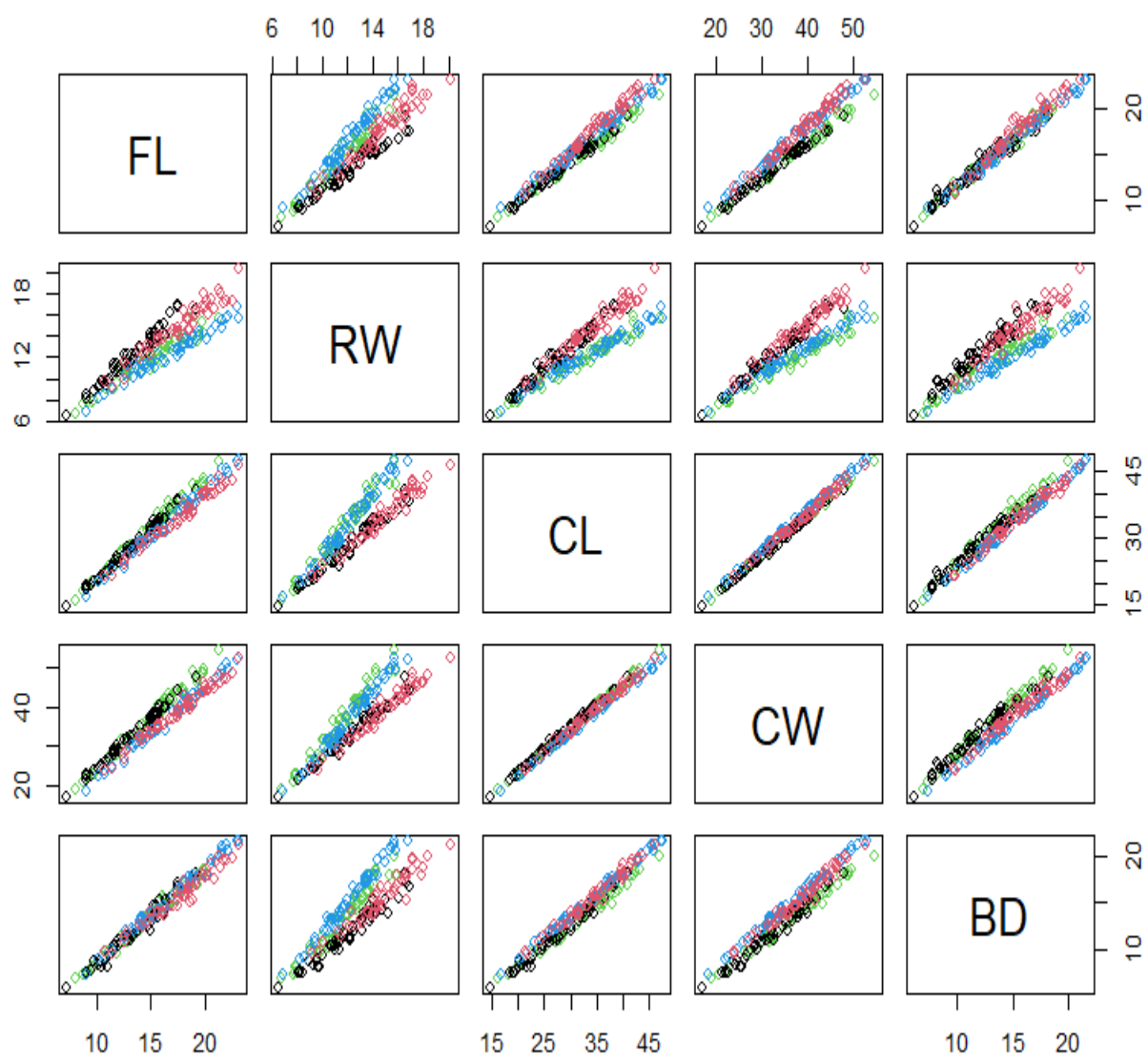
Mean Morphological Measurements by Species and Sex



5a. Produce a scatterplot matrix (pair plot) of five measurements

```
pairs(crabs[, c("FL", "RW", "CL", "CW", "BD")],  
      main = "Scatterplot Matrix of Crab Measurements",  
      col = as.numeric(interaction(crabs$sp, crabs$sex)))
```

Scatterplot Matrix of Crab Measurements



b. Describe the strength and nature of the relationships among variables

- Strength of Relationships

All five measurements are strongly positively correlated.

Larger crabs tend to be larger in all dimensions.

This indicates strong overall body size dependence.

- Nature of Relationships

Relationships appear approximately linear.

No obvious curvature or nonlinear patterns.

No clear clustering beyond group differences.

- Strongest Associations

The strongest relationships are typically between:

Carapace Length (CL) and Carapace Width (CW)

Rear Width (RW) and Carapace Width (CW)

These measure closely related aspects of overall shell size.

The variables largely measure overall body size, so:

```
> # They contain substantial redundant information.
```

```
> median_setosa_sepal_length <- median(iris$Sepal.Length[iris$Species == "setosa"])
```

```
> median_setosa_sepal_length
```

```
[1] 5
```

```
> mode_setosa_sepal_length <- iris$Sepal.Length[which.max(table(iris$Sepal.Length[iris$Species == "setosa"])]
```

```
>
```

